

Concepts and Algorithms for Multi-Criteria Route Optimization: A Comparison of Skyline Approaches

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1 Abstract

Modern navigation systems must increasingly optimize routes with respect to multiple, often conflicting criteria such as travel time, monetary cost, and energy consumption. Traditional approaches based on fixed weighting schemes are frequently too restrictive, as they cannot adequately represent diverse user preferences. Skyline-based methods address this limitation by identifying sets of non-dominated solutions, thereby providing multiple optimal alternatives for decision-making.

This paper compares three representative skyline-based route planning approaches: Advanced Route Skyline Computation (ARSC) [KRS10], Bicriteria Linear Route Skyline Computation (BLRSC) [SJSK14], and the Best-Compromise In-Route Nearest Neighbor model (BC-IRNN) [ACN17]. Although all three methods rely on graph-based search algorithms, they differ significantly in their optimization objectives and the definition of their result sets. ARSC computes the complete set of Pareto-optimal routes [KRS10], whereas BLRSC restricts the result set to linearly optimal paths, producing a more compact set of alternatives [SJSK14]. BC-IRNN extends these concepts to route planning scenarios that incorporate points of interest into an already planned route while minimizing both detour and overall travel distance [ACN17].

An analysis of the underlying concepts and experimental evaluations on real-world transportation networks reveals a clear progression across the examined approaches. While ARSC focuses on computing the complete Pareto skyline of routes, BLRSC improves both efficiency and usability by restricting the result set to linearly optimal routes. BC-IRNN further extends these concepts to in-route scenarios by integrating points of interest into an existing route, while balancing overall travel distance and detour costs. The comparison highlights the trade-off between completeness, result-set compactness, and computational efficiency, demonstrating how skyline-based routing has evolved from comprehensive route optimization towards more specialized and user-oriented decision-support systems.

2 Introduction

Traditional route planning algorithms typically optimize a single criterion, such as travel time or travel distance, and return one optimal route. In practice, however, routing decisions often involve multiple and potentially conflicting objectives. Besides minimizing travel time, users may also consider factors such as fuel consumption, travel costs, energy usage, environmental impact, or the number of transfers in public transportation systems. Since no single route is usually optimal with respect to all criteria, multi-criteria route planning has become an important research area.

A common approach to multi-criteria routing is the use of preference functions that combine several criteria into a single weighted objective. However, this requires users to specify precise weights in advance. In practice, such preferences are often difficult to quantify and may change depending on the situation. Consequently, fixed weighting schemes frequently fail to capture the diversity and dynamic nature of user preferences.

Skyline-based approaches address this limitation by identifying non-dominated alternatives instead of a single optimal route. A route belongs to the skyline if no other route performs at least as well in all criteria while being strictly better in at least one of them. As a result, skyline queries provide a set of meaningful alternatives that can satisfy a wide range of possible user preferences

without requiring these preferences to be specified beforehand.

Despite their advantages, skyline-based routing approaches face significant challenges. The number of possible paths between two locations grows exponentially with nodes in the network, and multi-criteria most optimal path problems are generally NP-hard. Furthermore, complete Pareto skylines often contain a large number of highly similar routes, making the result set difficult to interpret and reducing its practical usefulness. These challenges have motivated the development of advanced pruning and search-space reduction techniques that enable skyline computation on large real-world transportation networks.

This paper compares three representative approaches that address these challenges from different perspectives. Route Skyline Queries (BRSC and ARSC) [KRS10] compute the complete Pareto skyline of routes in multi-attribute networks and establish the foundation of skyline-based route planning. Linear Path Skyline Computation (BLRSC) [SJSK14] reduces the size of the result set by restricting it to linearly optimal routes in bicriteria networks. Best-Compromise In-Route Nearest Neighbor Queries (BC-IRNN) [ACN17] further extend skyline concepts to in-route scenarios, where users seek suitable points of interest while balancing overall travel distance and detour length.

By comparing the problem formulations, solution concepts, algorithmic strategies, and experimental results of these approaches, this work highlights the evolution of skyline-based route planning from complete Pareto optimization towards compact and application-specific recommendation systems. In particular, it illustrates the trade-offs between completeness, usability, user preference, and computational efficiency that characterize modern multi-criteria routing systems.

3 Paper Overview

3.1 Paper 1: Route Skyline Queries

The following section summarizes the route planning approach proposed by Kriegel, Renz, and Schubert [KRS10]. The authors introduce *Route Skyline Queries (RSQs)* as a novel solution for multi-criteria route planning that addresses the limitations of traditional routing algorithms. Conventional navigation systems typically optimize a single criterion or rely on fixed weight combinations of multiple criteria, requiring users to specify their preferences in advance. However, users are often unable to precisely quantify trade-offs between criteria such as travel time, distance, or fuel consumption.

To overcome this limitation, Route Skyline Queries compute the complete set of Pareto-optimal routes between a source and destination node. As a result, the returned solution set contains at least one optimal route for every possible positive weighting of the considered criteria, allowing users to select their preferred route without having to define their preferences beforehand.

The underlying data model is a *Multi-Attribute Network Graph (MAG)*, where each edge is associated with a vector of positive cost attributes such as travel distance, travel time, number of traffic lights, or fuel consumption. A route belongs to the skyline if it is not dominated by any other route and is therefore optimal for at least one possible user preference.

A major challenge is that the number of possible paths between two nodes can grow exponentially, making efficient skyline computation difficult. To address this issue, the authors combine graph-search techniques with graph embedding methods based on reference nodes (landmarks), enabling the computation of lower-bound estimates for multiple route attributes.

To compute route skylines, the paper introduces two algorithms: BRSC (Basic Route Skyline Computation) and ARSC (Advanced Route Skyline Computation). While BRSC relies on global pruning, ARSC extends the approach with local skyline pruning, allowing dominated subpaths to be removed earlier during the search process.

KRS10 establishes the foundation of skyline-based route planning, upon which both SJSK14 and ACN17 build.

3.2 Paper 2: Linear Path Skyline Computation in Bicriteria Networks

In their paper *Linear Path Skyline Computation in Bicriteria Networks* [SJSK14], Shekelyan, Jossé, Schubert, and Kriegel introduce a new approach to Route Skyline Queries called the *Linear*

Path Skyline and present an algorithm named *Bidirectional Linear Route Skyline Computation* (BLRSC).

The Linear Path Skyline differs from the conventional Route Skyline by imposing additional requirements for a path to be considered non-dominated. Besides conventional skyline dominance, a path can also be linearly dominated, if there exists no possible weighted sum of the criteria values under all reasonable user preferences, under which the benefit of the path in one criteria outweighs the trade-off in the other. As a result, more paths are considered dominated and fewer paths remain in the final non-dominated set. To implement this concept, the authors developed the BLRSC algorithm.

BLRSC uses the fact that every subpath of a non-dominated path must itself be non-dominated. Otherwise, a full-path differing only in that subpath would dominate the other path. The algorithm maintains a set of non-dominated subpaths ($S(s,v)$) from the source node (s) to every visited node (v).

Each newly generated subpath is first compared against the previously accepted candidate subpaths, using conventional dominance checks. To improve the efficiency of the subsequent linear dominance tests, the candidate subpaths are sorted according to their criteria values. This allows entire groups of subpaths to be discarded at once, once dominance against one path has been established.

Furthermore, a subpath is only considered part of the Linear Path Skyline, if it is not linearly dominated by any longer path crossing the same node, so this also has to be checked for. To further improve efficiency, the authors employ a reverse Dijkstra search for the computation of lower bounds during query processing.

Experiments on real-world datasets from Munich and Bavaria, as well as on randomly generated grids, show that BLRSC is approximately 60-80% faster at computing a Linear Path Skyline than ARSC [KRS10] is at computing a conventional Route Skyline.

3.3 Paper 3: Best-Compromise In-Route Nearest Neighbor

In the paper *Best-Compromise In-Route Nearest Neighbor Queries* [ACN17], Ahmadi, Costa, and Nascimento propose a new concept called Best-Compromise In-Route Nearest Neighbor (BC-IRNN). In a conventional IRNN query, the algorithm attempts to optimize a route with a detour to a nearby point of interest (POI), such as an ATM. Unlike conventional In-Route Nearest Neighbor queries, which solely minimize detour distance, BC-IRNN simultaneously optimizes two competing objectives: total travel distance $TD(P)$ and detour distance $DD(P, P^*)$. It therefore treats the problem as a bicriteria route skyline query and returns a set of non-dominated routes accordingly. It further applies the concept of linear dominance from SJSK14, to improve efficiency and reduce the size of the solution set to a more meaningful selection.

In the first phase, the shortest full path is computed using iterative nearest-neighbor POI retrieval together with an early stopping rule. Its detour distance provides an upper bound. In the second phase, the minimum-detour full path is computed. This path is approximated by selecting a POI near the preferred route using a group nearest-neighbor heuristic and restricting candidate POIs through an ellipse-based pruning strategy. Its travel distance provides an upper bound for the second optimization criterion.

Afterwards, each candidate path is checked for non-dominance against the previously accepted paths. If the candidate is not dominated, it is added to the result set, while any previously accepted paths dominated by the new candidate are removed.

Using real-world road networks from Amsterdam, Oslo, and Berlin, with varying user preferences and POI densities, the results show that the proposed method is orders of magnitude faster than a comparable baseline approach, with the performance gap increasing as route length grows. Furthermore, as POI density increases, the proposed method often becomes even faster, whereas the baseline slows down due to the larger number of POI candidates per subquery. The resulting linear skyline sets remain small, never exceeding ten paths across all tested configurations, demonstrating the paper’s goal of producing manageable and meaningful alternatives.

4 Comparison

4.1 Comparison of Problem Statements

Although all three papers address route skyline optimization, they focus on different routing problems.

While [KRS10] investigates the general problem of finding alternative routes between a source and a destination in a multi-attribute network, [SJSK14] focuses solely on bicriteria networks. And [ACN17] further modifies the problem by applying the concept of bicriteria networks to in-route nearest neighbor queries.

Focusing only on bicriteria problems can be useful, because the number of non-dominated solutions often grows rapidly with the number of criteria, making large skyline sets difficult to interpret in practical applications.

4.2 Comparison of Solution Concepts

[KRS10] is based on the concept of the *Pareto Skyline*. A route is included in the result set if it is not dominated by any other route. Consequently, the Route Skyline Query returns all routes that may be optimal for at least one possible user preference.

[SJSK14] addresses this limitation by introducing the concept of the *Linear Path Skyline*. Instead of returning the complete Pareto skyline, the approach only includes routes that are non-dominated even under all linear weighting of the considered criteria. As a result, the solution set becomes significantly smaller while still containing the most relevant alternatives for practical decision-making. In the operations research literature, such linearly optimal routes are commonly referred to as *supported solutions*.

[ACN17] adopts the same linear skyline principle, but applies it to in-route-nearest-neighbor queries.

4.3 Comparison of Algorithms and Search Strategies

[KRS10] introduces two algorithms for Route Skyline Queries: BRSC (Basic Route Skyline Computation) and ARSC (Advanced Route Skyline Computation). Both algorithms combine A*-based search with landmark-based graph embeddings to estimate lower bounds for multiple cost attributes. BRSC uses global pruning based on forward estimation, whereas ARSC introduces an additional local pruning strategy that removes dominated subpaths during the search. This significantly reduces both memory consumption and computational effort.

[SJSK14] builds directly upon the concepts introduced in KRS10. The proposed BLRSC algorithm also employs graph-search techniques and lower-bound estimations but exploits the geometric properties of bicriteria networks. Instead of searching for the complete Pareto skyline, BLRSC focuses exclusively on linearly optimal routes. This allows the algorithm to prune large parts of the search space and compute a much smaller result set more efficiently than ARSC.

[ACN17] also uses route expansion, skyline pruning, and best-compromise evaluation to efficiently identify relevant POIs along a preferred route.

4.3.1 Shared Foundations and Evolution

A clear chronological and conceptual dependency exists between the three approaches. All papers build upon classical graph-search algorithms, particularly Dijkstra’s algorithm and A*-based search techniques. Dijkstra’s algorithm is used both for shortest-path computation and for the generation of lower-bound estimates, while A* guides the search process through optimistic cost estimations.

[KRS10] provides the foundation by introducing Route Skyline Queries, the Multi-Attribute Network Graph model, and the BRSC and ARSC algorithms. The paper further establishes the use of graph embeddings and landmark-based lower-bound estimation for efficient skyline computation. Building upon this work, [SJSK14] identifies the limitations of complete Pareto skylines and introduces the concept of the Linear Path Skyline. Although the paper retains the skyline-search framework, it employs alternative lower-bound computation techniques and directly compares its BLRSC algorithm against ARSC.

Finally, [ACN17] builds upon both previous works. It adopts the concepts of linear dominance and linear skylines introduced by SJSK14 and applies them to the In-Route Nearest Neighbor problem. At the same time, it draws upon the route-skyline principles established in KRS10. Consequently, the three papers can be viewed as a progression of ideas: KRS10 establishes the theoretical and algorithmic foundation of route skylines, SJSK14 refines the concept through linear skylines, and ACN17 transfers these concepts to dynamic route-planning scenarios involving points of interest.

4.4 Comparison of Experimental Results

All three papers evaluate their proposed approaches on either real-world transportation networks or randomly generated datasets to demonstrate their effectiveness. Specifically, KRS10 compares the advanced ARSC algorithm with the simpler BRSC approach, SJSK14 evaluates BLRSC against ARSC, and ACN17 compares BC-IRNN with a baseline method.

The experiments presented in KRS10 demonstrate that the runtime difference between ARSC and BRSC increases exponentially as the number of hops increases. For paths of approximately 40 hops, BRSC requires more than ten times the runtime of ARSC. However, the runtime of ARSC also increases noticeably as the number of optimization criteria increases.

In the experiment of SJSK14, BLRSC consistently outperformed ARSC in terms of runtime by about 60-80%. While the performance advantage was particularly pronounced on smaller networks, the relative improvement decreased, as the size of the network increased.

Similarly, the evaluation presented in ACN17 shows that the runtime difference between BC-IRNN and a baseline approach increases exponentially with growing network size and increasing numbers of nodes. Importantly, BC-IRNN becomes even faster, when the number of points of interest (POIs) is increased within the same network.

4.5 Comparative Overview

Aspect	KRS10 (ARSC)	SJSK14 (BLRSC)	ACN17 (BC-IRNN)
Role in Development	Foundation of Route Skyline Queries	Extension of Route Skylines through Linear Skylines	Application of Linear Skylines to In-Route POI Queries
Builds Upon Problem	– General multi-criteria route planning	KRS10 Reduction of large skyline result sets	KRS10 and SJSK14 In-route nearest-neighbor routing with POIs
Objective	Complete Pareto-optimal routes	Compact set of linearly optimal routes	Best compromise between detour and travel distance
Criteria	Arbitrary number of criteria (d)	Two criteria (bicriteria)	Two criteria (travel distance and detour distance)
Result Set	Pareto Skyline	Linear Skyline	Linear Skyline with IRNN-detours
Core Algorithms Search Strategy	BRSC, ARSC A* + Graph Embedding	BLRSC Label Correction + Reverse Dijkstra Bounds	BC-IRNN A* + Elliptical Pruning
Main Innovation	Local skyline pruning	Linear dominance and skyline reduction	Detour-aware skyline search
Experimental Results	ARSC significantly outperforms BRSC, particularly for longer paths	BLRSC achieves approximately 60-80% lower runtime than ARSC while returning a smaller skyline set	BC-IRNN substantially outperforms a baseline approach and scales exceedingly well with increasing POI density
Strength	All potentially optimal routes	Compact and diverse alternatives	Optimal compromise for in-route stops
Limitation	Large result sets	Restricted to bicriteria networks	Specialized to preferred-path scenarios

Table 1: Evolution of skyline-based routing approaches from route skylines to in-route recommendation.

5 Conclusion

The comparison of three skyline-based approaches to multi-criteria route optimization (Route Skyline Queries (ARSC), Linear Path Skyline Computation (BLRSC), and Best-Compromise In-Route Nearest Neighbor Queries (BC-IRNN)) shows that although all three methods are based on skyline principles, the comparison shows clear differences in their objectives, solution concepts, and algorithmic implementations.

The analysis revealed a clear progression across the examined approaches. KRS10 focuses on computing the complete Pareto skyline of routes, thereby maximizing completeness and ensuring that all potentially optimal alternatives are returned. Building upon this foundation, SJSK14 introduces the Linear Path Skyline concept to address the practical limitations of large skyline result sets. By restricting the solution space to linearly optimal routes, the approach improves both interpretability and computational efficiency. Finally, ACN17 transfers these concepts to in-route scenarios, where users must balance overall travel distance and detour costs when visiting points of interest.

Across all three approaches, efficient search-space reduction plays a central role. Despite the computational complexity of multi-criteria routing problems, the proposed algorithms achieve practical performance through techniques such as skyline pruning, lower-bound estimation, graph embeddings, reverse Dijkstra searches, and elliptical pruning strategies. The experimental results demonstrate that these techniques enable skyline-based routing on large real-world transportation networks while maintaining manageable result sets.

Overall, the examined approaches demonstrate how skyline-based routing has evolved from the computation of complete Pareto-optimal route sets towards more specialized and user-oriented solutions. The comparison highlights the trade-off between completeness, usability, and computational efficiency, which remains a central challenge in multi-criteria route planning.

References

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